

Stability analysis of modular multilevel converter based on harmonic state-space theory

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Abstract: Modular multilevel converter (MMC) is regarded as one of the most potential topologies for high-voltage direct current transmission system. However, the multi-frequency behaviour and complex control structure of MMC bring great challenges to the stability analysis, identification of factors affecting stability, and controller design. To address this problem, the harmonic state-space (HSS) theory is applied to MMC in this study. First, the small-signal HSS model is established to describe the multi-frequency behaviour of MMC. Capacitor voltage control, circulating current control, and output current control are also taken into account in the modelling. On this basis, the stability analysis is performed by evaluating the eigenvalues of the HSS model, and participation factor is introduced to identify the factors affecting stability. In addition, the proposed approach reveals the influence of different controllers on the system stability, providing guidance for the tuning of multiple controller parameters. Finally, the effectiveness and accuracy of the proposed approach are verified by simulation and experiment results.

Nomenclature

$j = u, l$	arm
$k = a, b, c$	phase
u_{kg}/i_{kg}	output voltage/current
U_g/I_g	amplitude of output voltage/current
f_1	fundamental frequency
V_{dc}/i_{dc}	DC-side voltage/current
P	rated power
φ	power factor angle
C	submodule capacitance
C_{arm}	equivalent capacitance of arms
u_{kj}/i_{kj}	arm voltage/current
i_{kc}	circulating current
L_a	arm inductor
m_1	modulation ratio
N	submodule number per arm
R_a	equivalent arm resistor
u_{kcj}^i	capacitor voltage of the i th submodule
$u_{kcj} = \sum_{i=1}^N u_{kcj}^i$	total capacitor voltage of arms
$u_{kc\Delta} = u_{kcu} - u_{kcl}$	difference capacitor voltage between arms
$u_{kc\Sigma} = u_{kcu} + u_{kcl}$	total capacitor voltage of phases
S_{kj}	switching function of arms
$S_{k\Delta} = S_{ku} - S_{kl}$	differential-mode component of a switching function
$S_{k\Sigma} = S_{ku} + S_{kl}$	common-mode component of a switching function
T_s	sampling period

1 Introduction

The modular multilevel converter (MMC) has many attractive advantages, such as low harmonic distortion, high efficiency, modularity and scalability, etc. [1, 2]. All of these make MMC the most competitive solution for high-voltage and high power applications, including high-voltage direct current (HVDC) transmission, static synchronous compensation (STATCOM), large-scale energy storage system, etc. [3–5].

Owing to the unique structure of MMC, the arm energy fluctuates with multiple harmonic components. As a result, submodule capacitor voltages contain significant harmonic components, of which the fundamental and second harmonics are dominant. Also, even-order harmonic currents exist in the internal circulating current. These internal harmonics lead to complex frequency cross-coupling, strong non-linearity, and time-variance, making the modelling and analysis of MMC much more complicated than that of traditional voltage source converters [6]. In addition, there are multiple control objectives to be simultaneously achieved for both load requirements and stable operation of MMC, such as submodule capacitor voltage averaging and balancing, circulating current suppression, and output voltage or current regulation. The most commonly used control scheme of MMC is in a cascaded manner, consisting of multiple control loops and controllers [7]. Thus, how to achieve both the control performance and system stability is a great challenge, especially considering the internal harmonics of MMC.

In general, there are two main approaches for stability analysis of non-linear system: use non-linear analysis methods directly based on the non-linear model of the system, or linearise the system in order to use well known linear analysis techniques. In [8, 9], the open-loop and closed-loop stabilities of MMC were investigated by using Lyapunov theory. In [10], the stability analysis was directly performed based on time-periodic system theory (Poincare multipliers). However, non-linear methods have significant disadvantages, such as inconvenience for guiding controller design, complexity of analysis, and difficulty in practical application, etc. As a consequence, the widely used solution is to transform a time-varying model into a time-invariant model, perform small-signal linearisation, and then analyse the model using linear system theory. Based on this method, different linearised models were developed, and the stability analysis was performed based on eigenvalues analysis or Nyquist stability criterion. It is worth noting that MMC is a power converter with multi-frequency behaviour due to its significant harmonics in the circulating currents and capacitor voltages. Thus, a tradition modelling method widely used in 2-level VSC cannot be directly applied to MMC. In [11–13], multi-harmonic linearisation method was proposed to build the impedance model of MMC. This method deals with multi-harmonics by transforming the model from time domain into the frequency domain. Then all the harmonic

components can be expressed by the Fourier coefficients. Although, this method presents a good approach to linearise the multi-harmonics topology, it is complex and not intuitive [14]. To address this problem, harmonic state-space (HSS) theory [15–17] was applied to MMC for impedance modelling [18, 19]. The HSS model is easy to be implemented by computer and to be extended to any number of harmonics for high accuracy. Nevertheless, the impedance-based method does not consider the stability of internal circulating current and capacitor voltage. In addition, it is inconvenient to identify the factors that affect system stability and guide the controllers design. Furthermore, eigenvalues-based methods were used to investigate the small-signal stability of MMC in the dq frame [20–23]. The small-signal model contains multiple dq frames so as to have the internal harmonics to be considered. However, this method involves multiple dq frames and lengthy algebra; the complexity of the analysis increases rapidly with the increase of considered harmonics.

The stability analysis approach proposed in this paper is based on HSS theory. In this approach, the small-signal HSS model of MMC is developed to describe the frequency cross coupling of different state variables. Then the stability analysis is performed by evaluating the eigenvalues of the HSS model. Participation factor is introduced to identify the correspondence between state variables and eigenvalues. On this basis, the proposed approach is able to develop insight into the stability zones of multiple controllers, and, therefore, can have significant theoretical and practical importance. What distinguish this paper from the contribution in [18, 19] is that the proposed approach takes the internal stability of circulating current and capacitor voltage into account, and can easily identify the factors leading to instability, providing better guidance for the design of multiple controllers.

The rest of this paper is organised as follows. The fundamental principles of HSS theory and the steady-state analysis of MMC are presented in Section 2. On this basis, the linearised small-signal HSS model of MMC is developed in Section 3. Then, the stability analysis is performed by evaluating the eigenvalues of the HSS model in Section 4. In Section 5, the effectiveness and accuracy of proposed stability analysis method are verified by simulation and experiments results. Finally, conclusions are presented in Section 6.

2 HSS-based steady-state analysis of MMC

2.1 HSS formulation

The linear time-periodic (LTP) system's dynamic equation is expressed as

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (1)$$

It is well known that any continuous periodic function can be expressed in the form of Fourier series. Thus, each state variable $x(t)$ can be expressed as

$$x(t) = e^{st} \sum_{k \in \mathbb{Z}} X_k e^{jk\omega_1 t} \quad (2)$$

where $\omega_1 = 2\pi f_1$; e^{st} is multiplied to represent the transient process of the variable; X_k is the Fourier coefficient of the k th harmonic, which is obtained by

$$X_k = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} x(t) e^{-jk\omega_1 t} dt \quad (3)$$

Likewise, the input variable can be rewritten as the Fourier series form

$$u(t) = e^{st} \sum_{k \in \mathbb{Z}} U_k e^{jk\omega_1 t} \quad (4)$$

The matrices $A(t)$ and $B(t)$ are coefficients of the state-space model, which contain no transient process. Hence, they can be

directly rewritten as the form of Fourier series without multiplying e^{st} .

$$\begin{aligned} A(t) &= \sum_{k \in \mathbb{Z}} A_k e^{jk\omega_1 t} \\ B(t) &= \sum_{k \in \mathbb{Z}} B_k e^{jk\omega_1 t} \end{aligned} \quad (5)$$

Substituting (2)–(5) into (1), the state-space equations can be conveniently written as follows [16, 17]:

$$sX = (A - N)X + BU \quad (6)$$

where

$$X = [X_{-h}^T \quad \dots \quad X_{-1}^T \quad X_0^T \quad X_1^T \quad \dots \quad X_h^T]^T \quad (7)$$

$$U = [U_{-h}^T \quad \dots \quad U_{-1}^T \quad U_0^T \quad U_1^T \quad \dots \quad U_h^T]^T \quad (8)$$

$$\begin{aligned} A &= \Gamma(A_{-h}, \dots, A_{-1}, A_0, A_1, \dots, A_h) \\ &= \begin{bmatrix} A_0 & A_{-1} & \dots & A_{-h} & & & \\ A_1 & \ddots & \ddots & \ddots & \ddots & & \\ \vdots & \ddots & A_0 & A_{-1} & \ddots & \ddots & \\ A_h & \ddots & A_1 & A_0 & A_{-1} & \ddots & A_{-h} \\ & \ddots & \ddots & A_1 & A_0 & \ddots & \vdots \\ & & \ddots & \ddots & \ddots & \ddots & A_{-1} \\ & & & A_h & \dots & A_1 & A_0 \end{bmatrix} \end{aligned} \quad (9)$$

$$N = \begin{bmatrix} -jh\omega_1 I & & & & & & \\ & \ddots & & & & & \\ & & 0I & & & & \\ & & & \ddots & & & \\ & & & & -jh\omega_1 I & & \end{bmatrix} \quad (10)$$

$$B = \Gamma(B_{-h}, \dots, B_{-1}, B_0, B_1, \dots, B_h) \quad (11)$$

Matrices A and B are Toeplitz matrices, in which all the elements on the diagonal are the same. To simplify the expression, the Toeplitz matrix is denoted by the function $\Gamma()$. It's noted that the elements of A and B are still matrices having the same size with $A(t)$ and $B(t)$, respectively. N is a diagonal matrix that represents the frequency information; I is an identity matrix that has the same matrix size with the number of state variables. Assuming that the number of state variables in $x(t)$ is n , the number of state variables in (6) is $(2h+1)n$. In general, the infinitely dimensioned matrices in the HSS form can be truncated to a reasonably finitely dimensioned matrices by choosing an appropriate h .

2.2 Steady-state analysis of MMC

The three-phase MMC is illustrated in Fig. 1. It consists of three phases, each with two arms. An arm is composed of N series-connected submodules, an arm inductor, and an equivalent arm resistor. Submodule may use a half-bridge or a full-bridge topology, which, however, does not affect the discussion here.

Note that this paper is devoted to study MMC on the system level, thus, it is assumed an average model and ideal balancing between SM_j in one arm. Then, each arm can be treated as one equivalent submodule with an equivalent capacitor $C_{arm} = C/N$ and equivalent capacitor voltage u_{kcj} . On this basis, the model of MMC can be described using four state variables: output current, circulating current, total capacitor voltage of phases and imbalance capacitor voltage between arms. The state-space form of the MMC model is [17, 18]

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (12)$$

where

$$x(t) = [i_{kc}(t) \quad i_{kg}(t) \quad u_{kc\Sigma}(t) \quad u_{kc\Delta}(t)]^T \quad (13)$$

$$u(t) = [V_{dc} \quad u_{kg}(t)]^T \quad (14)$$

$$A(t) = \begin{bmatrix} -\frac{R_a}{L_a} & 0 & -\frac{S_{k\Sigma}(t)}{4L_a} & \frac{S_{k\Delta}(t)}{4L_a} \\ 0 & -\frac{R_a}{L_a} & -\frac{S_{k\Delta}(t)}{2L_a} & -\frac{S_{k\Sigma}(t)}{2L_a} \\ \frac{S_{k\Sigma}(t)}{C_{arm}} & \frac{S_{k\Delta}(t)}{2C_{arm}} & 0 & 0 \\ \frac{S_{k\Delta}(t)}{C_{arm}} & \frac{S_{k\Sigma}(t)}{2C_{arm}} & 0 & 0 \end{bmatrix} \quad (15)$$

$$B(t) = \begin{bmatrix} 1/2L_a & 0 & 0 & 0 \\ 0 & -2/L_a & 0 & 0 \end{bmatrix}^T \quad (16)$$

Ignoring the switching frequency harmonics, switching functions of arms mainly consist of DC, fundamental and second harmonic components:

$$\begin{aligned} S_{ku} &= \frac{1}{2}m_0 - \frac{1}{2}m_1\sin(\omega_1 t + \theta_1) - \frac{1}{2}m_2\sin(\omega_2 t + \theta_2) \\ S_{kl} &= \frac{1}{2}m_0 + \frac{1}{2}m_1\sin(\omega_1 t + \theta_1) - \frac{1}{2}m_2\sin(\omega_2 t + \theta_2) \end{aligned} \quad (17)$$

It's noted that there are complex frequency cross-coupling among different state variables due to the harmonic components in the circulating current and capacitor voltages. To describe the frequency cross-coupling accurately, the HSS modelling method is applied to MMC. Following the modelling procedure in (1)–(6), the time-domain state-space model is transformed into the HSS form. The detailed expressions of the matrices can be directly obtained, which are not listed again.

In steady-state condition, it is obvious that

$$(A - N)X + BU = 0 \quad (18)$$

When the MMC adopts open-loop control structure, the switching functions are directly given. Then, the steady-state values of the state variables can be obtained as

$$X_s = (A - N)^{-1}(BU) \quad (19)$$

Under the closed-loop control structure, the switching functions are affected by the state variables across the feedback. As a result, (18) becomes a non-linear equation where the quadratic non-linearity is caused by the product of state variables from A and X . Hence, the steady-state values cannot be directly solved by (19). In this case, non-linear solving method has to be used. One example is given here to show how to obtain the steady-state values.

One generalised control scheme of MMC is presented in Fig. 2, consisting of capacitor voltage control, circulating current control, and output current control. Capacitor voltage control is divided into two parts: capacitor voltage averaging control and capacitor voltage balancing control. The former balances the average capacitor voltage of all arms by regulating the DC and fundamental circulating currents. $G_f(s)$ is the filter to prevent capacitor voltage ripples from affecting the circulating current. PI_1 and PI_2 are proportional–integral (PI) controllers to eliminate the error of $u_{kc\Sigma}$ and $u_{kc\Delta}$, respectively. The latter ensures capacitor voltage balance within arms by selecting appropriate submodules according to their capacitor voltages. Circulating current and output current controls are performed in stationary reference frame ($\alpha\beta$) through proportional–resonant (PR) controllers.

In steady-state condition, the DC component of capacitor voltage, circulating current, and output current equal to their references, respectively,

$$\begin{aligned} i_{kg} &= I_{kg}\cos(\omega_1 t + \varphi) \\ i_{kc} &= P/3V_{dc} \\ u_{kcu}(dc) &= u_{kcu}(dc) = V_{dc} \end{aligned} \quad (20)$$

Then, the elements of the matrix X are

$$\begin{aligned} X_0 &= [P/3V_{dc} \quad 0 \quad 2V_{dc} \quad 0]^T \\ X_{\pm 1} &= \begin{bmatrix} 0 & \frac{1}{2}I_g e^{\pm j\varphi} & \frac{1}{2}U_{kc\Sigma 1} e^{\pm j\alpha_1} & \frac{1}{2}U_{kc\Delta 1} e^{\pm j\gamma_1} \end{bmatrix}^T \\ X_{\pm h} &= \begin{bmatrix} 0 & 0 & \frac{1}{2}U_{kc\Sigma h} e^{\pm j\alpha_h} & \frac{1}{2}U_{kc\Delta h} e^{\pm j\gamma_h} \end{bmatrix}^T, h \geq 2 \end{aligned} \quad (21)$$

where $U_{kc\Sigma h}$, α_h and $U_{kc\Delta h}$, γ_h are the amplitude and phase angle of the h th harmonic component of $u_{kc\Sigma h}$ and $u_{kc\Delta h}$, respectively.

Defining a non-linear function F :

$$F(p) = (A - N)X + BU \quad (22)$$

where p denotes the variables to be solved:

$$p = [m_0, m_1, \theta_1, m_2, \theta_2, U_{kc\Sigma 1}, \alpha_1, \dots, U_{kc\Sigma h}, \alpha_h, U_{kc\Delta 1}, \gamma_1, \dots, U_{kc\Delta h}, \gamma_h] \quad (23)$$

Finally, solving the non-linear equation $F(p) = 0$ yields the steady-state values of p . A typical non-linear equation solving method is the 'fsolve' function in Matlab. As an example, with the main

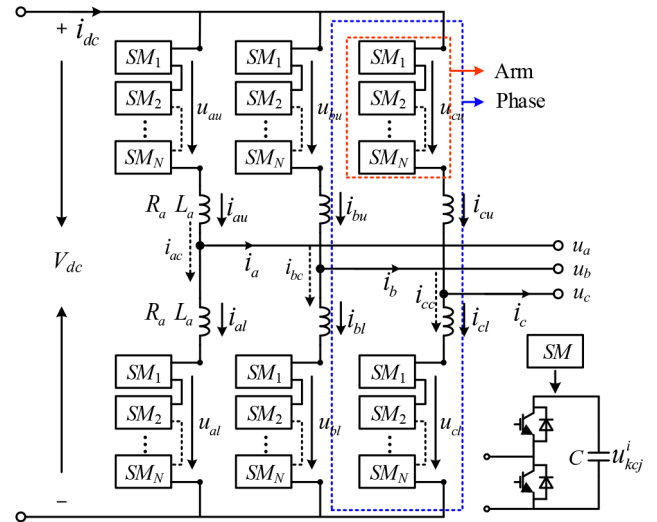


Fig. 1 Structure of three-phase MMC

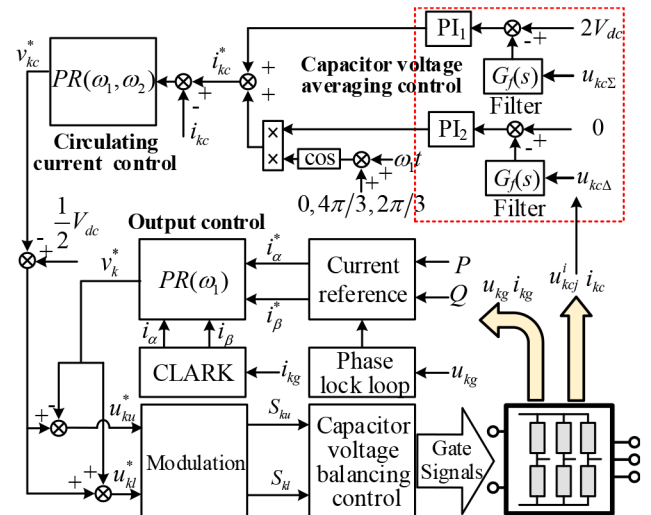


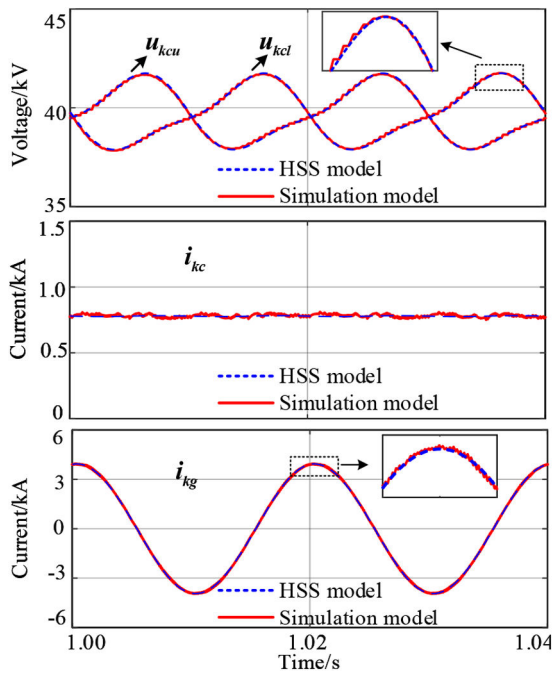
Fig. 2 Control structure of MMC

Table 1 Main circuit parameters of simulation model

Parameters	Symbol	Value
DC voltage	V_{dc}	40 kV
AC volatile	U_g	16 kV
frequency	f_1	50 Hz
submodule voltage	V_{sm}	2.5 kV
submodule number per arm	N	16
submodule capacitor	C	20 mF
arm inductor	L_a	5 mH
submodule resistor	R_a	0.1 Ω
rated power	P	96 MW
power factor angle	φ	0

Table 2 Solved switching functions

Power, MW	m_0	m_1	θ_1	m_2	θ_2
48	0.9981	0.8036	0.0678	0.0151	1.6200
96	0.9944	0.8094	0.1354	0.0302	1.6686

**Fig. 3** Comparison between HSS model and time-domain nonlinear simulation model

parameters in Table 1, the solved switching functions are listed in Table 2.

To verify the accuracy of the HSS modelling method, the comparison between HSS model and time-domain non-linear simulation model has been carried out. Substituting the switching functions in Table 2 into (18), the steady-state values of state variables can be obtained. The considered harmonic order is $h = 2$. The frequency-domain expression is transformed into time-domain expression through inverse Fourier transform, and the results are presented in Fig. 3. The simulation results are obtained through the time-domain nonlinear simulation model in the Matlab/Simulink. It can be seen that the results of HSS model are in agreement well with the simulation model (in Matlab/Simulink), indicating the HSS modelling method is able to describe the frequency cross coupling of MMC accurately.

3 Small-signal HSSM of MMC

In this section, the HSS modelling is performed in two separate parts. First, small-signal analysis method is applied to the time-domain model in (12), and the small-signal of the topology part is obtained in time domain. Second, the small-signal model of the

control part in Fig. 2 is established. Nevertheless, it has to be mentioned that the proposed stability analysis approach is not only limited to the control scheme in Fig. 2. In the following analysis, the subscript ‘ p ’ and ‘ c ’ denote the topology part and control part, respectively; the subscript ‘ s ’ denotes the steady-state operating point; Δ denotes small-signal perturbation.

3.1 Modelling of topology part

The time-domain state-space model of MMC is presented in (12). Injecting small-signal perturbations and ignoring high-order terms, the small-signal model of the topology part is expressed as

$$\Delta \dot{x}_p(t) = A_p(t)\Delta x_p(t) + B_p(t)\Delta u_p(t) \quad (24)$$

where

$$\Delta x_p(t) = [\Delta i_{kc}(t) \quad \Delta i_{kg}(t) \quad \Delta u_{kc\Sigma}(t) \quad \Delta u_{kc\Delta}(t)]^T \quad (25)$$

$$\Delta u_p(t) = [\Delta S_{k\Sigma} \quad \Delta S_{k\Delta} \quad \Delta V_{dc} \quad \Delta u_{kg}]^T \quad (26)$$

$$A_p(t) = \begin{bmatrix} -\frac{R_a}{L_a} & 0 & -\frac{S_{k\Sigma}(t)}{4L_a} & \frac{S_{k\Delta}(t)}{4L_a} \\ 0 & -\frac{R_a}{L_a} & -\frac{S_{k\Delta}(t)}{2L_a} & -\frac{S_{k\Sigma}(t)}{2L_a} \\ \frac{S_{k\Sigma}(t)}{C_{arm}} & \frac{S_{k\Delta}(t)}{2C_{arm}} & 0 & 0 \\ \frac{S_{k\Delta}(t)}{C_{arm}} & \frac{S_{k\Sigma}(t)}{2C_{arm}} & 0 & 0 \end{bmatrix} \quad (27)$$

$$B_p(t) = \begin{bmatrix} -\frac{u_{kc\Sigma s}(t)}{4L_a} & \frac{u_{kc\Delta s}(t)}{4L_a} & \frac{1}{2L_a} & 0 \\ -\frac{u_{kc\Delta s}(t)}{2L_a} & -\frac{u_{kc\Sigma s}(t)}{2L_a} & 0 & -\frac{2}{L_a} \\ \frac{i_{kcs}(t)}{C_{arm}} & \frac{i_{kgs}(t)}{2C_{arm}} & 0 & 0 \\ \frac{i_{kgs}(t)}{2C_{arm}} & \frac{i_{kcs}(t)}{C_{arm}} & 0 & 0 \end{bmatrix} \quad (28)$$

3.2 Modelling of control part

Capacitor voltage ripples only contain the integer multiples of the fundamental frequency. Hence, the moving average filter tuned at the fundamental frequency is adopted as the capacitor voltage filter, whose transfer function is

$$G_f(s) = \frac{1 - e^{-sT_1}}{sT_1} \quad (29)$$

The first-order Pade approximation of the MAF is a first-order low-pass filter (LPF). There is a great difference between them in the frequency range higher than the corner frequency of LPF [24]. For high accuracy, the second-order Pade approximate is applied to (29):

$$G_f(s) = \frac{1}{1 + (1/2)T_1s + (1/12)T_1^2s^2} \quad (30)$$

Then the small-signal model of the MAF is described by the following differential equations:

$$\frac{d\Delta u_{kc\Sigma_f}}{dt} = \Delta x_1 \quad (31)$$

$$\frac{d\Delta x_1}{dt} = -\frac{12}{T_1^2}\Delta u_{kc\Sigma_f} - \frac{6}{T_1}\Delta x_1 + \frac{12}{T_1^2}\Delta u_{kc\Sigma}$$

$$\frac{d\Delta u_{kc\Delta_f}}{dt} = \Delta x_2 \quad (32)$$

$$\frac{d\Delta x_2}{dt} = -\frac{12}{T_1^2}\Delta u_{kc\Delta_f} - \frac{6}{T_1}\Delta x_2 + \frac{12}{T_1^2}\Delta u_{kc\Delta}$$

where $u_{kc\Sigma_f}$ and $u_{kc\Delta_f}$ are the filtered capacitor voltages of upper and lower arms, respectively; x_1 and x_2 are the differentiation of them, respectively.

In Fig. 2, capacitor voltage averaging control is performed by regulating the DC and fundamental components of circulating current. Then the resulted perturbation in circulating current is

$$\Delta i_{kc}^*(t) = K_{pv1}\Delta u_{kc\Sigma_f}(t) + K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1t + K_{iv1}\Delta x_3(t) + K_{iv2}\Delta x_4(t)\cos\omega_1t \quad (33)$$

(see (34))

(see (35))

(see (36))

where K_{pv1} , K_{iv1} and K_{pv2} , K_{iv2} are the proportional and integral coefficients of PI controllers PI₁ and PI₂, respectively; x_3 and x_4 are defined as

$$\begin{aligned} \frac{d\Delta x_3(t)}{dt} &= -\Delta u_{kc\Sigma_f}(t) \\ \frac{d\Delta x_4(t)}{dt} &= \Delta u_{kc\Delta_f}(t) \end{aligned} \quad (37)$$

It is worth noting that the resonant part of the PR controller only affects the controlled variable around the designated frequency while having little impact at other frequencies. Hence, only the proportional gain of the PR controller is considered in the following analysis. Then the perturbations of the arm voltage references are obtained as

$$\begin{aligned} \Delta u_{ku}^*(t) &= K_{pc}\Delta i_{kc}(t) - K_{po}\Delta i_{kg}(t) \\ &\quad - K_{pc}K_{iv1}\Delta x_3(t) - K_{pc}K_{iv2}\Delta x_4(t)\cos\omega_1t \\ &\quad + K_{pc}K_{pv1}\Delta u_{kc\Sigma_f}(t) \\ &\quad - K_{pc}K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1t \end{aligned} \quad (38)$$

$$\begin{aligned} \Delta u_{kl}^*(t) &= K_{pc}\Delta i_{kc}(t) + K_{po}\Delta i_{kg}(t) \\ &\quad - K_{pc}K_{iv1}\Delta x_3(t) - K_{pc}K_{iv2}\Delta x_4(t)\cos\omega_1t \\ &\quad + K_{pc}K_{pv1}\Delta u_{kc\Sigma_f}(t) \\ &\quad - K_{pc}K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1t \end{aligned} \quad (39)$$

where K_{po} and K_{pc} are the proportional gains of output current PR controller and circulating current PR controller, respectively.

The unit computation delay and the ZOH are also considered, and its continuous time-domain transfer function is

$$G_d(s) = e^{-sT_s} \frac{1 - e^{-sT_s}}{sT_s} \simeq e^{-1.5sT_s} \quad (40)$$

Applying first-order Pade approximation to (40), the control delay is simplified to

$$G_d(s) = \frac{1 - (3/4)T_s s}{1 + (3/4)T_s s} \quad (41)$$

Then the perturbations of the switching functions are

$$\begin{aligned} \Delta S_{ku}(t) &= \Delta x_5(t) - \frac{\Delta u_{ku}^*(t)}{V_{dc}} \\ \Delta S_{kl}(t) &= \Delta x_6(t) - \frac{\Delta u_{kl}^*(t)}{V_{dc}} \end{aligned} \quad (42)$$

$$\Delta x_c(t) = [\Delta x_1(t) \quad \Delta x_2(t) \quad \Delta x_3(t) \quad \Delta x_4(t) \quad \Delta x_5(t) \quad \Delta x_6(t) \quad \Delta u_{kc\Sigma_f}(t) \quad \Delta u_{kc\Delta_f}(t)]^T \quad (34)$$

$$A_c(t) = \begin{bmatrix} -\frac{6}{T_1} & 0 & 0 & 0 & 0 & 0 & -\frac{12}{T_1^2} & 0 \\ 0 & -\frac{6}{T_1} & 0 & 0 & 0 & 0 & 0 & -\frac{12}{T_1^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{pv1} & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{iv2} \cos\omega_1t & -\frac{4}{3T_s} & 0 & \frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{pv1} & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{pv2} \cos\omega_1t \\ 0 & 0 & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{iv1} & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{iv2} \cos\omega_1t & 0 & -\frac{4}{3T_s} & \frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{pv1} & -\frac{8}{3T_s} \frac{K_{pc}}{V_{dc}} K_{pv2} \cos\omega_1t \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (35)$$

$$C_c(t) = \begin{bmatrix} 0 & 0 & \frac{2K_{pc}}{V_{dc}} K_{iv1} & \frac{2K_{pc}}{V_{dc}} K_{iv2} \cos\omega_1t & 1 & 1 & -\frac{2K_{pc}}{V_{dc}} K_{pv1} & \frac{2K_{pc}}{V_{dc}} K_{pv2} \cos\omega_1t \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \end{bmatrix} \quad (36)$$

where x_5 and x_6 are defined as

$$\begin{aligned}\frac{d\Delta x_5(t)}{dt} &= -\frac{4}{3T_s}\Delta x_5(t) + \frac{8}{3T_s}\frac{\Delta u_{ku}^*(t)}{V_{dc}} \\ \frac{d\Delta x_6(t)}{dt} &= -\frac{4}{3T_s}\Delta x_6(t) + \frac{8}{3T_s}\frac{\Delta u_{kl}^*(t)}{V_{dc}}\end{aligned}\quad (43)$$

Equations (31), (32), (37) and (43) can be reorganised as the state-space form. Then, the small-signal model of the control part is

$$\begin{aligned}\Delta \dot{x}_c(t) &= A_c(t)\Delta x_c(t) + B_c(t)\Delta u_c(t) \\ \Delta y_c(t) &= C_c(t)\Delta x_c(t) + D_c(t)\Delta u_c(t)\end{aligned}\quad (44)$$

where

$$\Delta u_c(t) = [\Delta i_{kc}(t) \quad \Delta i_{kg}(t) \quad \Delta u_{k\sigma}(t) \quad \Delta u_{k\Delta}(t)]^T \quad (45)$$

$$B_c(t) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{8}{3T_s}\frac{K_{pc}}{V_{dc}} & \frac{8}{3T_s}\frac{K_{pc}}{V_{dc}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8}{3T_s}\frac{K_{pc}}{V_{dc}} & \frac{8}{3T_s}\frac{K_{po}}{V_{dc}} & 0 & 0 \\ \frac{12}{T_1^2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12}{T_1^2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (46)$$

$$\Delta y_c(t) = [\Delta S_{k\sigma}(t) \quad \Delta S_{k\Delta}(t)]^T \quad (47)$$

$$D_c(t) = \begin{bmatrix} -\frac{2K_{pc}}{V_{dc}} & 0 & 0 & 0 \\ 0 & \frac{2K_{po}}{V_{dc}} & 0 & 0 \end{bmatrix} \quad (48)$$

3.3 Small-signal model of the whole system

The small-signal model of MMC is presented in Fig. 4 by connecting the model of topology part and control part. Then the small-signal model of the whole system is expressed as

$$\begin{aligned}\begin{bmatrix} \Delta \dot{x}_c(t) \\ \Delta \dot{x}_p(t) \end{bmatrix} &= \begin{bmatrix} A_c(t) & 0 \\ 0 & A_p(t) \end{bmatrix} \begin{bmatrix} \Delta x_c(t) \\ \Delta x_p(t) \end{bmatrix} \\ &+ \begin{bmatrix} B_c(t) & 0 \\ 0 & B_p(t) \end{bmatrix} \begin{bmatrix} \Delta u_c(t) \\ \Delta u_p(t) \end{bmatrix}\end{aligned}\quad (49)$$

It is noted that the input variables of the topology part contain the output variables of the control part, then (49) is rewritten as

$$\begin{aligned}\begin{bmatrix} \Delta \dot{x}_c(t) \\ \Delta \dot{x}_p(t) \end{bmatrix} &= \begin{bmatrix} A_c(t) & 0 \\ 0 & A(t) \end{bmatrix} \begin{bmatrix} \Delta x_c(t) \\ \Delta x_p(t) \end{bmatrix} \\ &+ \begin{bmatrix} B_c(t) & 0 & 0 \\ 0 & B_{p1}(t) & B_{p2}(t) \end{bmatrix} \begin{bmatrix} \Delta u_c(t) \\ \Delta u_{p1}(t) \\ \Delta u_{p2}(t) \end{bmatrix}\end{aligned}\quad (50)$$

where $B_{p1}(t)$ is the first two columns of the matrix $B_p(t)$ and $B_{p2}(t)$ is the last two columns of $B_p(t)$; $u_{p1}(t)$ and $u_{p2}(t)$ are the first and last two rows of the input matrix $u_p(t)$, respectively.

In addition, the following relations are found:

$$\begin{aligned}\Delta u_{p1}(t) &= \Delta y_c(t) = C_c(t)\Delta x_c(t) + D_c(t)\Delta u_c(t) \\ \Delta u_c(t) &= \Delta x_p(t)\end{aligned}\quad (51)$$

Substituting (51) into (50) yields the time-domain small-signal model of the whole system

$$\begin{aligned}\begin{bmatrix} \Delta \dot{x}_c(t) \\ \Delta \dot{x}_p(t) \end{bmatrix} &= \begin{bmatrix} A_c(t) & B_c(t) \\ B_{p1}(t)C_c(t) & A_p(t) + B_{p1}(t)D_c(t) \end{bmatrix} \begin{bmatrix} \Delta x_c(t) \\ \Delta x_p(t) \end{bmatrix} \\ &+ \begin{bmatrix} 0 \\ B_{p2}(t) \end{bmatrix} \Delta u_{p2}(t)\end{aligned}\quad (52)$$

Finally, the time-domain small-signal model can be directly transformed into the HSS form, according to (1)–(6). The detailed expressions of the matrices would not be listed here again, and the small-signal HSS model of MMC is written as

$$s\Delta X = (A - N)\Delta X + B\Delta U \quad (53)$$

4 Stability analysis of MMC

The small-signal HSS model of MMC is obtained in Section 3, then the stability analysis can be performed by evaluating the eigenvalues of the matrix $A - N$. This method can be used to investigate the influence of different factors on stability, such as main circuit parameters, control structure, and controller parameters. Due to the limited space, this section only takes the controller parameters as an example to show how to use the proposed method.

In order to investigate the interactions among different controllers and provide theoretical guidance for the design of multiple controllers, the correspondence between controllers and stability should be identified first. In this paper, the harmonic order considered is $h = 2$, and there are total $(2h + 1)n = 60$ state variables in (53). To identify which states contribute to the different oscillation modes, the participation factor analysis is introduced in this paper [25].

The right and left modal matrices of $A - N$ are defined as

$$\begin{aligned}\Phi &= [\phi_1 \quad \phi_2 \quad \cdots \quad \phi_i] \\ \Psi &= [\psi_1^T \quad \psi_2^T \quad \cdots \quad \psi_i^T]^T\end{aligned}\quad (54)$$

where ϕ_i and ψ_i are the right and left eigenvectors of the matrix $A - N$ associated with eigenvalue λ_i . They can be calculated by

$$\begin{aligned}(A - N)\phi_i &= \lambda_i\phi_i \\ \psi_i(A - N) &= \lambda_i\psi_i\end{aligned}\quad (55)$$

The participation factor is defined as

$$p_{ki} = \Phi_{ki}\Psi_{ik} \quad (56)$$

The participation factor p_{ki} is a measure of the relative participation of the k th state variable in the i th mode, and vice versa. The analysis results of different oscillation modes at the rated operating conditions are summarised in Table 3, where state variables with a participation ratio higher than 0.1 are all listed. In Table 3, oscillation modes are divided into eight groups according to their main participating state variables.

The oscillation modes in Group I include $\lambda_{1,2}$, $\lambda_{3,5}$, $\lambda_{8,9}$, $\lambda_{11,14}$, and $\lambda_{16,18}$, corresponding to the state variable i_{kg} at frequency $\pm 2f_1$, $\pm f_1$

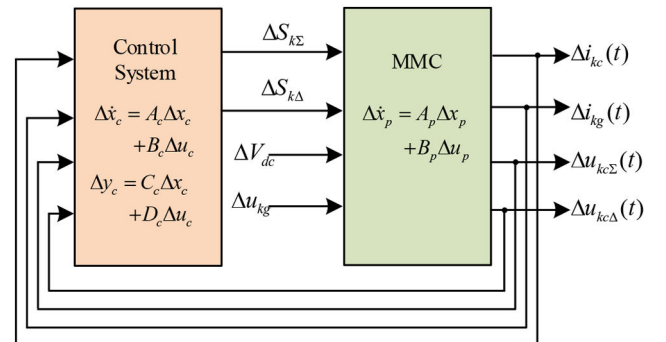


Fig. 4 Small-signal model of the whole system

and 0 (DC component), respectively. The dynamic performance of output current control is mainly decided by these oscillation modes. And the root loci with varying output current controller proportional gain K_{po} is presented in Fig. 5a. It can be seen that the oscillation modes in Group I move towards the right half plane with the increase of controller gain K_{po} , while the other groups nearly remain unchanged. Then, the stability of output current will be deteriorated, and when $K_{po} = 7.0$, output current becomes instable.

Likewise, Group II consists of oscillation modes $\lambda_{4,6}$, $\lambda_{19,20}$, $\lambda_{7,10}$, $\lambda_{15,17}$ and $\lambda_{12,13}$, corresponding to the second harmonic, fundamental and DC component of circulating current, respectively. Fig. 5b shows the eigenvalue loci with varying circulating current

controller proportional gain K_{pc} . Group II moves toward to the right half plane with the increase of K_{pc} , while the other eigenvalues nearly remain unchanged. When K_{pc} increases to 14.0, circulating current becomes instable.

The other groups are related to capacitor voltage control. Group III and Group V are only related to the control of total capacitor voltage $u_{kc\Sigma}$; Group IV and Group VI are only related to the control of imbalance capacitor voltage $u_{kc\Delta}$. It has to be noted that Group VII and Group VIII are related to both $u_{kc\Sigma}$ and $u_{kc\Delta}$, indicating that the capacitor voltage controllers PI₁ and PI₂ are influenced by each other. To guarantee the stability of capacitor voltage, they should be uniformly designed.

Table 3 Eigenvalues, oscillation modes, and main participation state variables of the system ($K_{pv1} = 0.05$, $K_{pv2} = 0.05$, $K_{pc} = 6.0$, $K_{po} = 4.0$)

Group	Oscillation modes	Eigenvalues	Damping factor	Participating state variables	Group	Oscillation modes	Eigenvalues	Damping factor	Participating state variables
group I	$\lambda_{1,2}$	$-524.9 \pm j2629.3$	0.20	$i_{kg} (\pm 2f_1)$	group II	$\lambda_{4,6}$	$-719.7 \pm j2256.7$	0.30	$i_{kc} (\pm 2f_1)$
	$\lambda_{3,5}$	$-525.7 \pm j2324.7$	0.22	$i_{kg} (\pm f_1)$		$\lambda_{7,10}$	$-720.1 \pm j1944.5$	0.34	$i_{kc} (\pm f_1)$
	$\lambda_{8,9}$	$-526.1 \pm j2009.6$	0.25	$i_{kg} (0)$		$\lambda_{12,13}$	$-720.6 \pm j1628.9$	0.40	$i_{kc} (0)$
	$\lambda_{11,14}$	$-526.1 \pm j1695.3$	0.31	$i_{kg} (\pm f_1)$		$\lambda_{15,17}$	$-721.1 \pm j1314.7$	0.48	$i_{kc} (\pm f_1)$
	$\lambda_{16,18}$	$-527.5 \pm j1377.8$	0.36	$i_{kg} (\pm 2f_1)$		$\lambda_{19,20}$	$-720.1 \pm j1003.5$	0.57	$i_{kc} (\pm 2f_1)$
group III	$\lambda_{23,31}$	$-180.5 \pm j627.2$	0.26	$u_{kc\Sigma}, x_1 (\pm 2f_1)$	group IV	$\lambda_{26,34}$	$-45.3 \pm j622.5$	0.07	$u_{kc\Delta}, x_2 (\pm 2f_1)$
	$\lambda_{39,47}$	$-180.5 \pm j314.2$	0.50	$u_{kc\Sigma}, x_1 (\pm f_1)$		$\lambda_{40,48}$	$-54.9 \pm j314.2$	0.17	$u_{kc\Delta}, x_2 (\pm f_1)$
	λ_{53}	-180.5	1.00	$u_{kc\Sigma}, x_1(0)$		λ_{58}	-54.8	1.00	$u_{kc\Sigma}, x_2(0)$
group V	$\lambda_{24,33}$	$-23.0 \pm j628.8$	0.04	$u_{kc\Sigma_f}, x_3 (\pm 2f_1)$	group VI	$\lambda_{25,32}$	$-16.6 \pm j627.8$	0.03	$u_{kc\Delta_f}, x_4 (\pm 2f_1)$
	$\lambda_{42,50}$	$-23.0 \pm j314.4$	0.07	$u_{kc\Sigma_f}, x_3 (\pm f_1)$		$\lambda_{41,49}$	$-17.5 \pm j314.2$	0.06	$u_{kc\Delta_f}, x_4 (\pm f_1)$
group VII	λ_{59}	-23.0	1.00	$u_{kc\Sigma_f}, x_3(0)$	group VIII	λ_{60}	-17.5	1.00	$u_{kc\Delta_f}, x_4(0)$
	$\lambda_{21,29}$	$-70.2 \pm j710.6$	0.10	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm 2f_1)$		$\lambda_{22,30}$	$-133.9 \pm j700.8$	0.19	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm 2f_1)$
	$\lambda_{27,36}$	$-66.9 \pm j544.2$	0.12	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm 2f_1)$		$\lambda_{28,35}$	$-138.6 \pm j559.2$	0.24	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm 2f_1)$
	$\lambda_{38,46}$	$-68.5 \pm j395.1$	0.17	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm f_1)$		$\lambda_{37,45}$	$-131.8 \pm j381.1$	0.33	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm f_1)$
	$\lambda_{44,52}$	$-68.6 \pm j230.3$	0.28	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm f_1)$		$\lambda_{43,51}$	$-131.7 \pm j246.3$	0.47	$u_{kc\Sigma_f}, u_{kc\Delta_f} (\pm f_1)$
	$\lambda_{55,57}$	$-68.5 \pm j83.9$	0.63	$u_{kc\Sigma_f}, u_{kc\Delta_f}(0)$		$\lambda_{54,56}$	$-131.8 \pm j67.5$	0.89	$u_{kc\Sigma_f}, u_{kc\Delta_f}(0)$

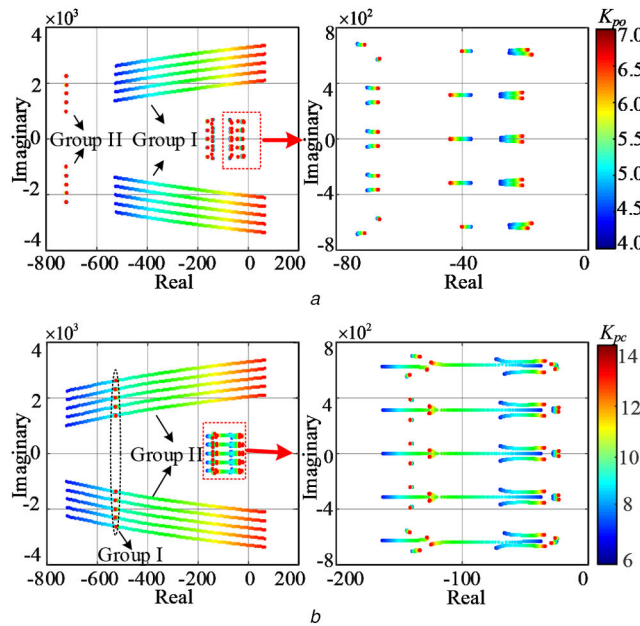


Fig. 5 Influence of output current and circulating current controllers on stability
(a) Root loci with varying output current controller gain K_{po} ,
(b) Root loci with varying circulating current controller gain K_{pc}

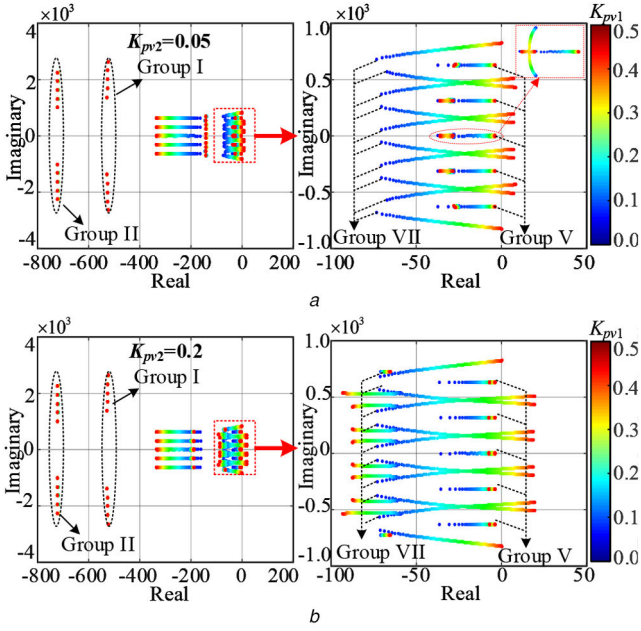


Fig. 6 Influence of capacitor voltage proportional gain K_{pv1} on stability
(a) Root loci with varying capacitor voltage controller gain K_{pv1} when $K_{pv2} = 0.05$,
(b) Root loci with varying capacitor voltage controller gain K_{pv1} when $K_{pv2} = 0.20$

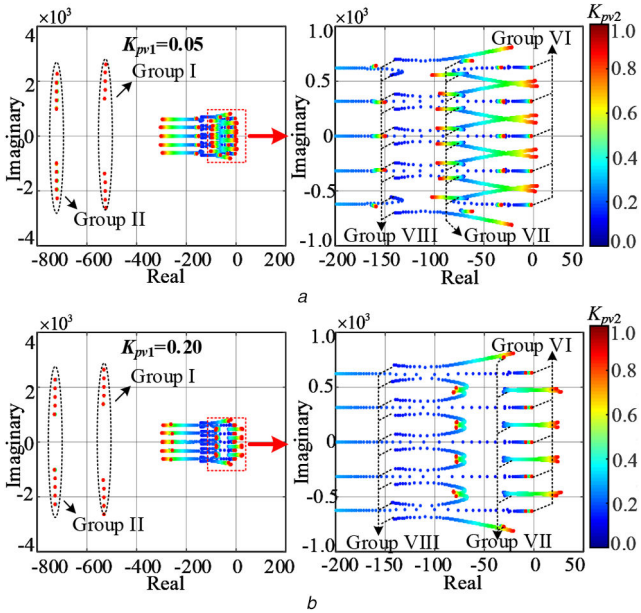


Fig. 7 Influence of capacitor voltage proportional gain K_{pv2} on stability;
(a) Root loci with varying capacitor voltage controller gain K_{pv2} when $K_{pv1} = 0.05$,
(b) Root loci with varying capacitor voltage controller gain K_{pv2} when $K_{pv1} = 0.20$

For further investigation of the interaction between capacitor voltage controllers PI_1 and PI_2 , Figs. 6–8 present the eigenvalue loci with varying controller proportional gains K_{pv1} , K_{pv2} , and integral gains K_{iv1} , K_{iv2} , respectively. In Fig. 6a, Group I and Group II remain unchanged with the increasing of K_{pv1} . Group III moves towards to the left side, which does not affect the stability. Group V and Group VII move towards to the right half plane, and the system become unstable when $K_{pv1} = 0.4$. In Fig. 6b, K_{pv2} increases to 0.2. It can be seen that Group VII moves faster than that in Fig. 6a. When K_{pv1} increases to 0.3, the system becomes unstable. The same conclusion can also be obtained in Fig. 7. In Fig. 7a, $K_{pv1} = 0.05$, Group VIII locates at the right half plane when $K_{pv2} = 0.90$. However, when $K_{pv1} = 0.20$, Group VIII will not move to the right half plane, and the instability is caused by Group VII. In Fig. 8, with the increasing of integral gain, the dominant eigenvalues move towards to the left side and the stability increases. Then the

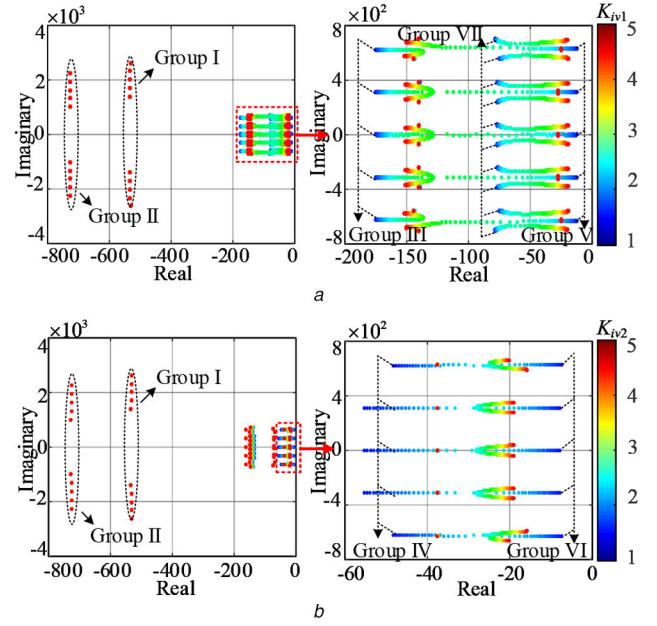


Fig. 8 Influence of integral gain K_{iv1} and K_{iv2} on stability
(a) Root loci with varying integral coefficient K_{iv1} ,
(b) Root loci with varying a integral coefficient K_{iv2}

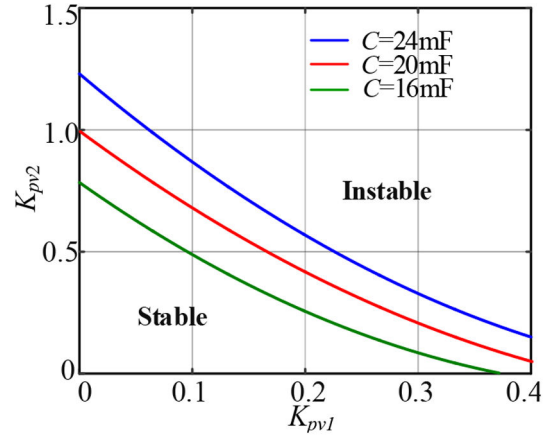


Fig. 9 Relation between capacitor voltage proportional gains K_{pv1} and K_{pv2}

dominant eigenvalues move towards to the right side and the stability decreases. Hence, the stability can be improved by optimising the design of controller parameters. At last, with the main circuit parameters in Table 1, the relation between K_{pv1} and K_{pv2} are shown in Fig. 9. It can be seen that K_{pv1} is in negative correlation to K_{pv2} for different submodule capacitors. It is worth noting that when one controller gain K_{pv1} (K_{pv2}) exceeds some limit, the system would be instable no matter how to design the other one controller gain K_{pv2} (K_{pv1}).

5 Verification

In this section, analytical results based on eigenvalue analysis of the HSS model, the simulations of the time-domain non-linear simulation model, and the experimental results are presented. The validation of the proposed approach will focus on the stability threshold value of controller gains, including the output current controller, circulating current controller and capacitor voltage controller.

5.1 Simulation results

The parameters of the simulation model are the same as Table 1. According to the eigenvalues analysis in Fig. 5, output current becomes instable when controller gain K_{po} increases to 7.0. The

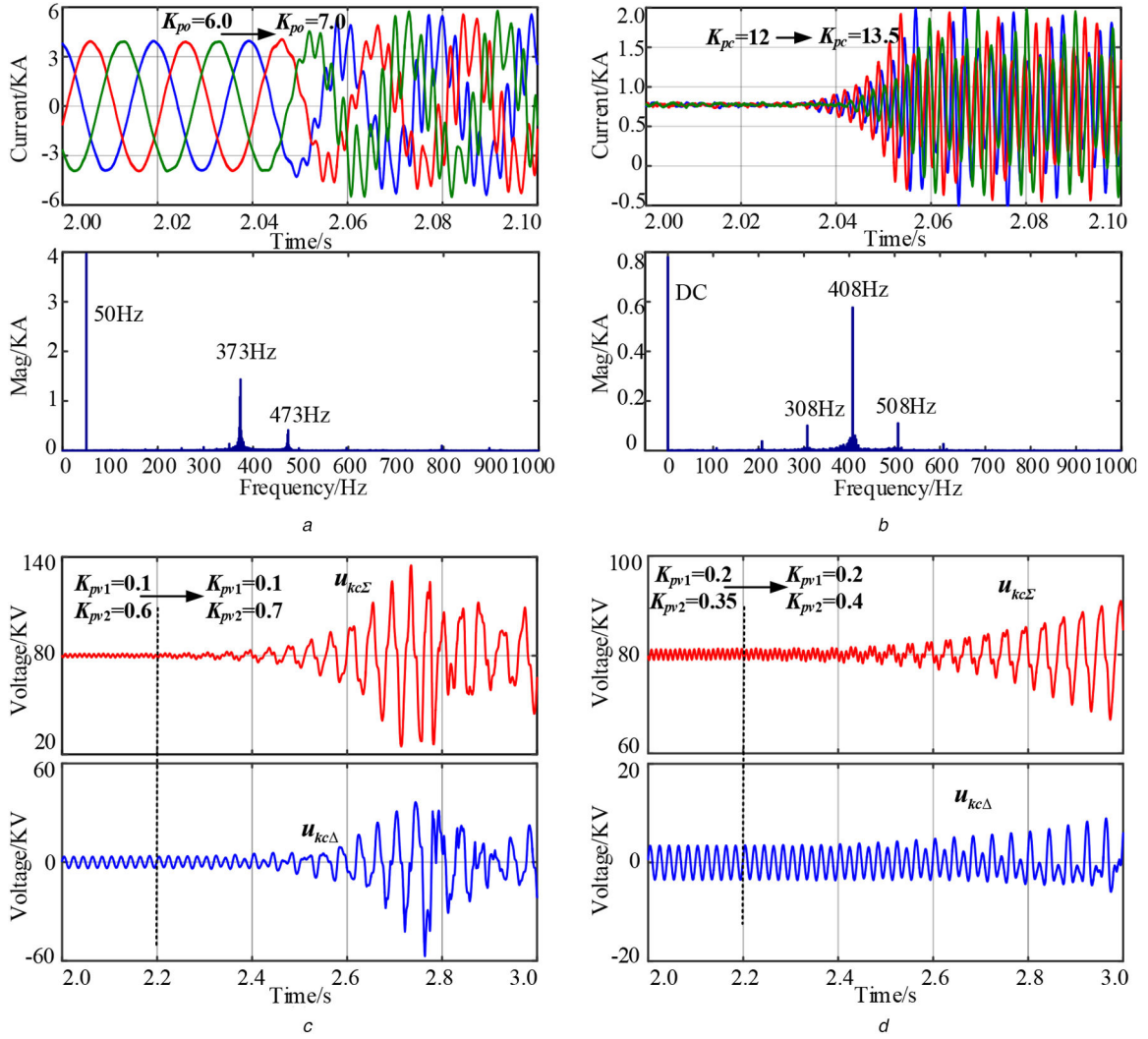


Fig. 10 Simulation results of stability analysis

- (a) Output current,
(b) Circulating current,
(c) Capacitor voltage when $K_{pv1} = 0.1$, and K_{pv2} steps to 0.70 from 0.60,
(d) Capacitor voltage when $K_{pv1} = 0.2$, and K_{pv2} steps to 0.40 from 0.35

eigenvalues in the right half plane are $24.7 \pm j2070$, $24.7 \pm j2387$, $25.1 \pm j2702$, $25.5 \pm j3016$, and $25.9 \pm j3326$, corresponding to output current at frequency $-2f_1$, $-f_1$, 0 , f_1 and $2f_1$, respectively. The time-domain simulation result is illustrated in Fig. 9a. At time $T = 2.04$, K_{po} steps to 7.0 from 6.0, then the output current becomes instable. The FFT analysis shows that the oscillations at frequency 373 and 473 Hz are dominant, corresponding to oscillation modes $24.7 \pm j2387$ and $25.5 \pm j3016$, respectively. In Fig. 7b, circulating current becomes unstable when circulating current controller gain K_{pc} steps to 13.5. The eigenvalues in the right half plane are $8.6 \pm j2020$, $8.5 \pm j2375$, $8.3 \pm j2700$, $8.2 \pm j3010$, and $7.5 \pm j3320$, corresponding to the circulating current at frequency $-2f_1$, $-f_1$, 0 , f_1 , and $2f_1$, respectively. Circulating current mainly consists of DC and second harmonics, hence the oscillation modes $64.3 \pm j2109$, $65.1 \pm j2741$, and $65.7 \pm j3365$ are dominant, which have been clearly verified by the FFT analysis.

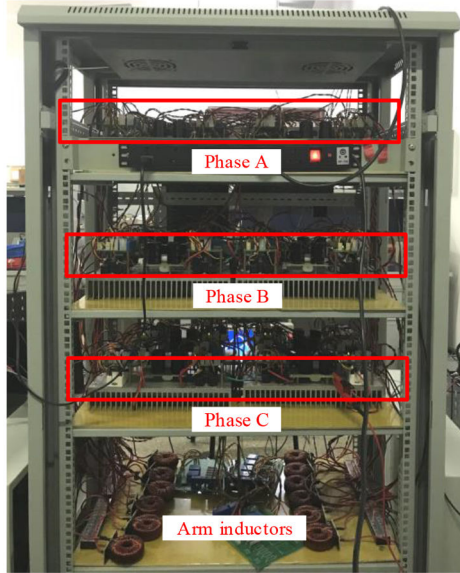
Capacitor voltage balancing control among phases and between arms are not independently, and the relation of K_{pv1} and K_{pv2} has been presented in Fig. 9. As an example, the simulation results when $K_{pv1} = 0.1$ and $K_{pv1} = 0.2$ are illustrated in Fig. 10. It is obvious that the stability analysis proposed in this paper can accurately identify the stability of the system.

5.2 Experiment results

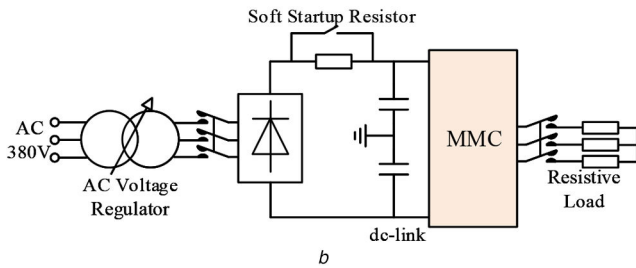
A downscaled three-phase MMC prototype was built in the laboratory as shown in Fig. 11 and the main circuit parameters are listed in Table 4. The control structure of the prototype was designed with a single digital-signal processor (DSP) as the main controller and three field-programmable gate arrays (FPGAs) as the auxiliary controllers in each phase. Owing to the limitation of experimental conditions, each arm only consists of two submodules, and the ac side adopts open-loop control schemes with a resistance load $R_{load} = 5\Omega$. However, it does not affect the proposed stability analysis approach. Following the analysis in Section 4, the stability boundary of different controllers is listed in Table 5. With the increase of K_{pv1} , the stability boundary of K_{pv2} decreases, while circulating current controller gain K_{pc} is not affected. The experimental results of the stability boundary validation are illustrated in Fig. 12. It is obvious that the experiment results are in agreement well with the analytical results listed in Table 4. Hence, both the simulation and experimental results verify the accuracy of the proposed stability analysis approach.

6 Conclusion

In this paper, a general method based on HSS theory is proposed to perform a stability analysis of MMC. First, the accurate steady-state operating point of MMC is obtained based on HSS theory.



a



b

Fig. 11 Configuration of the three-phase MMC prototype

(a) Picture of the prototype,
(b) Circuit of the prototype

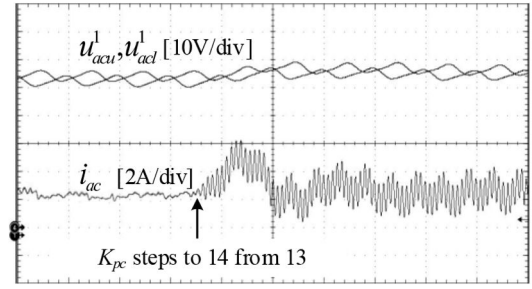
Table 4 Main circuit parameters of experiment prototype

Parameters	Symbol	Value
DC voltage	V_{dc}	120 V
AC volatile	U_g	48 V
frequency	f_1	50 Hz
submodule voltage	V_{sm}	60 V
submodule number per arm	N	2
submodule capacitor	C	5 mF
arm inductor	L_a	5 mH
submodule resistor	R_a	0.05 Ω
load	R_{load}	5 Ω

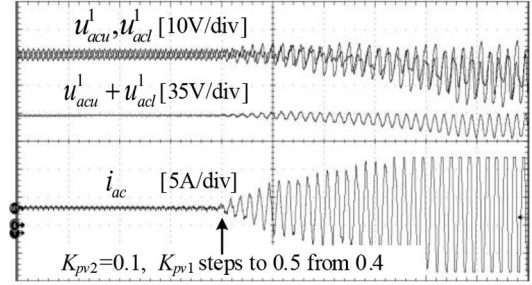
Table 5 Stability boundary of controller gains

K_{pv1}	K_{pv2}	K_{pc}
0.1	0.80	13.5
0.2	0.55	13.5
0.3	0.35	13.5
0.4	0.15	13.5
0.5	0.02	13.5

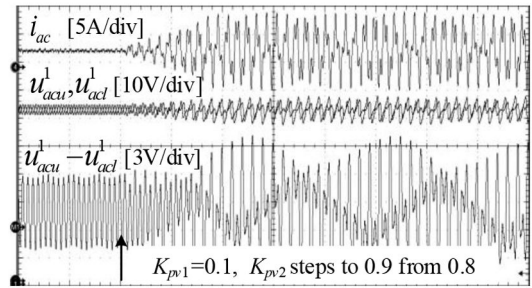
Then, the small-signal HSS model is established to characterise the multi-frequency behaviour of MMC, in which capacitor voltage control, circulating current control, and output current control are considered. Eigenvalue analysis is applied to the HSS model for stability analysis, and participation factor is introduced to identify which state variables contribute different oscillation modes. On this basis, the proposed approach can develop insight into the stability zones of multiple controllers. What's more, the proposed method can be used to study the effects of other factors on stability, such as main circuit parameters and control structure, and, therefore, can



a



b



c

Fig. 12 Experiment results of stability analysis

(a) K_{pc} steps to 14 from 13,
(b) $K_{pv2} = 0.1$, K_{pv1} steps to 0.5 from 0.4,
(c) $K_{pv1} = 0.1$, K_{pv2} steps to 0.9 from 0.8

have significant theoretical and practical importance. Finally, the stability analysis approach is verified by both simulation and experiment results.

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